



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
GOVERNMENT HYDRO ENGINEERING COLLEGE  
BANDLA (BILASPUR)

## G-HYDRO HLM

A WEB-BASED HOSTEL LEAVE MANAGEMENT SYSTEM FOR GHEC

CAPSTONE PROJECT - CSEE-611P

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# INTRODUCTION:



## 1. Background

In hostels, the leave process is handled manually using paper forms & registers.

This traditional method is slow, inefficient, and difficult to maintain.

## 2. Current Scenario

Students must fill out forms and submit them physically to the warden. The approval process takes time and lacks proper tracking.

## 3. Proposed System

G-Hydro HLM is a web-based hostel leave management system. It allows students to apply for leave online and get approvals digitally.

## 4. Benefits

The system reduces paperwork, saves time, and improves data accuracy. It also provides a centralized platform for easy access and management.

Government Hydro Engineering college, Bandia, Bilaspur H.P.  
Hostel leave form Dated: \_\_\_\_\_  
I, \_\_\_\_\_ D/O \_\_\_\_\_ is a \_\_\_\_\_  
Student (Branch) (Year) Roll no. \_\_\_\_\_ of GHEC, Bilaspur resides in room no. \_\_\_\_\_ of the Girls Hostel. I may be permitted to leave the Hostel on date \_\_\_\_\_ from (Date/Time) \_\_\_\_\_

|                               |         |                      |                                              |
|-------------------------------|---------|----------------------|----------------------------------------------|
| Name of the parents/ Guardian | Address | Signature of student | Whether informed to parents/ Guardian or not |
| Mobile No. _____              |         | Mobile No. _____     | YES/NO                                       |

Signature of Warden \_\_\_\_\_



GHEC Hostel

Welcome Back

Sign in to your account

Student Superintendent Warden

Chief Warden Director

Roll No. or Email

Enter your credentials

Password

Remember me Forgot password?

Sign In →

Don't have an account? Register here >



# PROBLEM STATEMENT :

## 1. Manual Leave Process

The existing hostel leave system is paper-based and requires physical form submission. This makes the process slow and difficult to manage.

## 2. Lack of Proper Tracking

There is no centralized system to monitor leave requests or check their status. Students and wardens face difficulty in tracking records.

## 3. High Chances of Errors

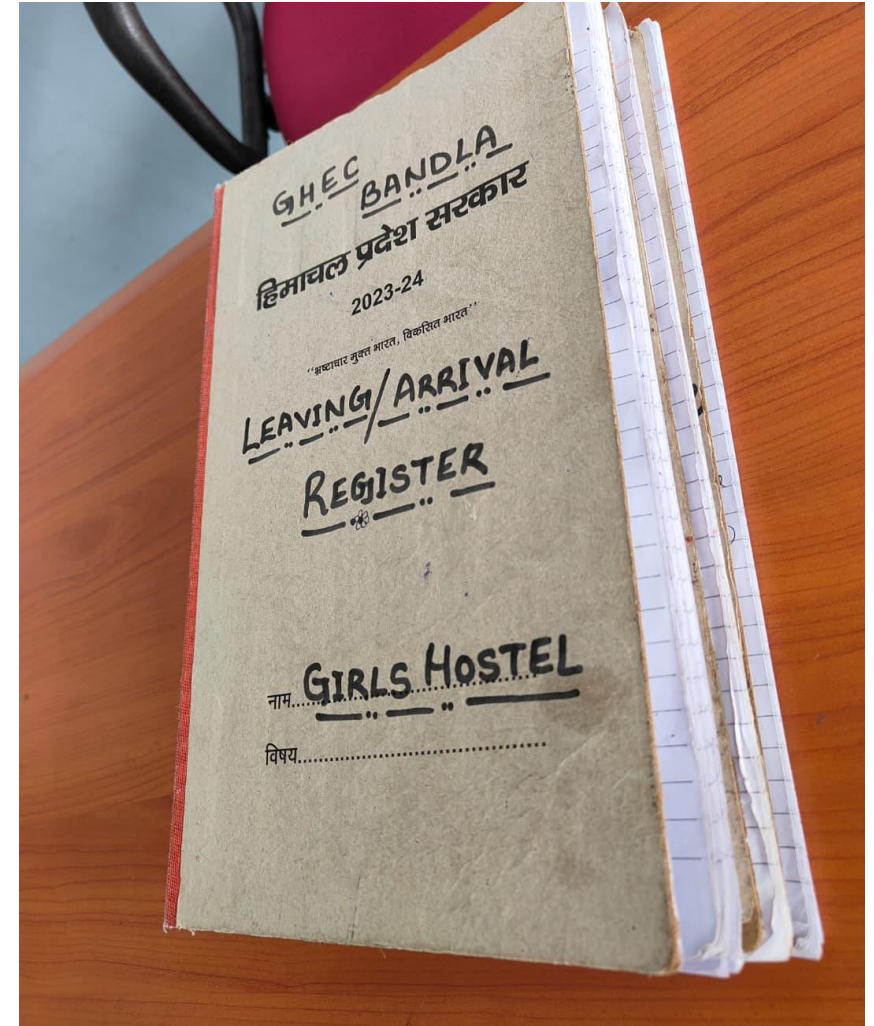
Manual data entry can lead to mistakes, loss of records, and mismanagement. It also creates confusion in maintaining accurate data.

## 4. Delay in Approval

Leave approvals depend on manual verification by wardens, which takes time. This causes unnecessary delays for students.

## 5. Need for Digital Solution

There is a need for a web-based system that automates the leave process, ensures accuracy, provides real-time tracking, and reduces paperwork.



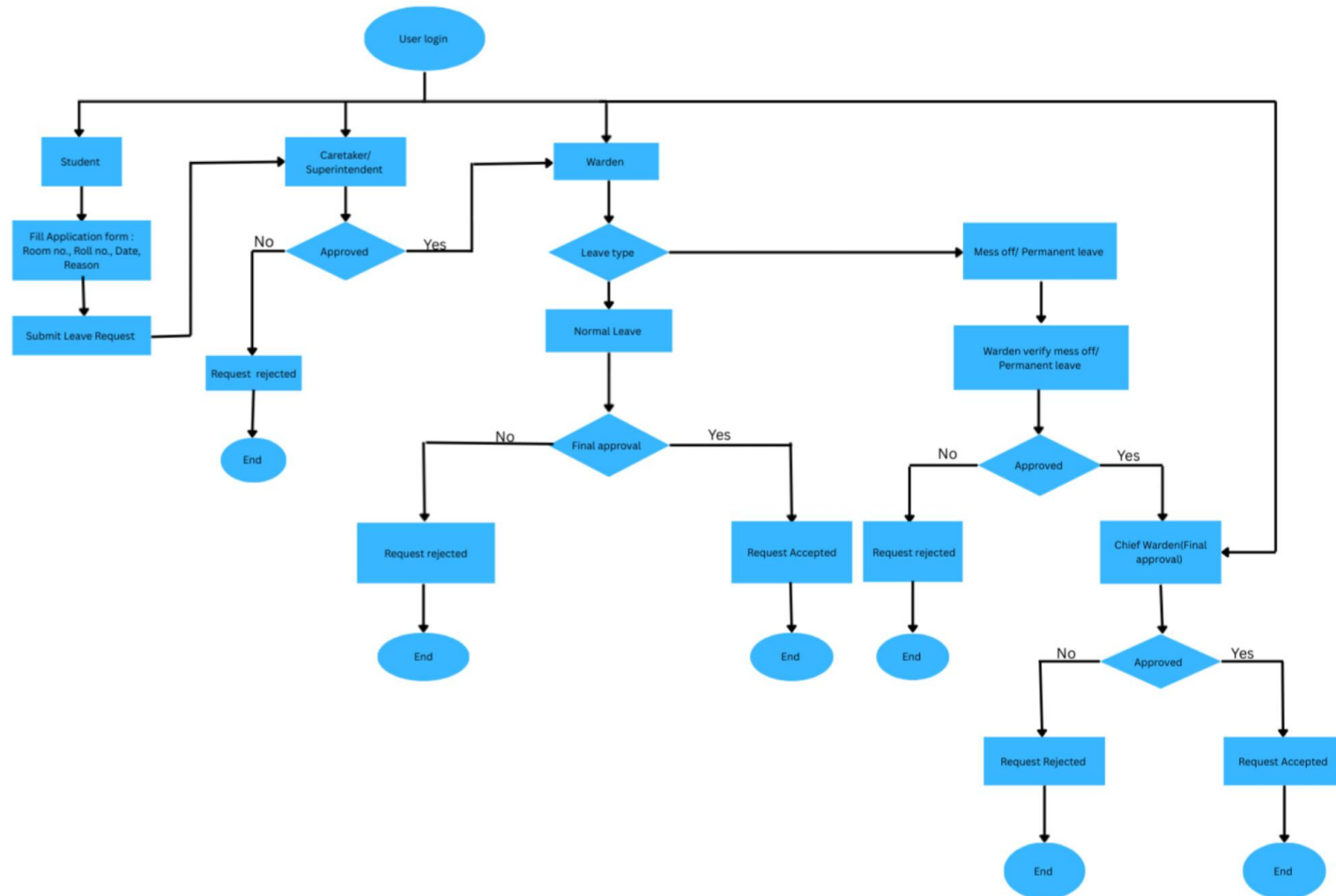
# METHODOLOGY:

## 1. Requirement Analysis

- Identify problems in the current manual leave system
- Collect requirements from students, wardens, and admin
- Define system features like login, leave request, and approval

## 2. System design

- Design overall architecture of the web-based system
- Create database structure (students, leave records, approvals)
- Plan user interface (simple and user-friendly screens)



# METHODOLOGY:



## 3. Development

- **Frontend:** Forms for leave application
- Designed user interface using HTML, CSS, JavaScript
- Created pages like:
  - Login/Register
  - Apply Leave Form
  - Leave Status Dashboard
- **Backend :** Implemented server-side logic using python
- Handled:
  - User authentication (login/signup)
  - Leave request processing
  - Approval/rejection by warden
- **Database Integration**
- Used database (e.g., MySQL)
- Stored and retrieved data efficiently

### Apply for Leave

Fill out the details below

Mess Off Quota

Balance: 5 / 10 Days

Leave Type

Contact No.

Select Type

+9198765 43210

From Date

To Date

Reason for Leave

Explain why you need leave...

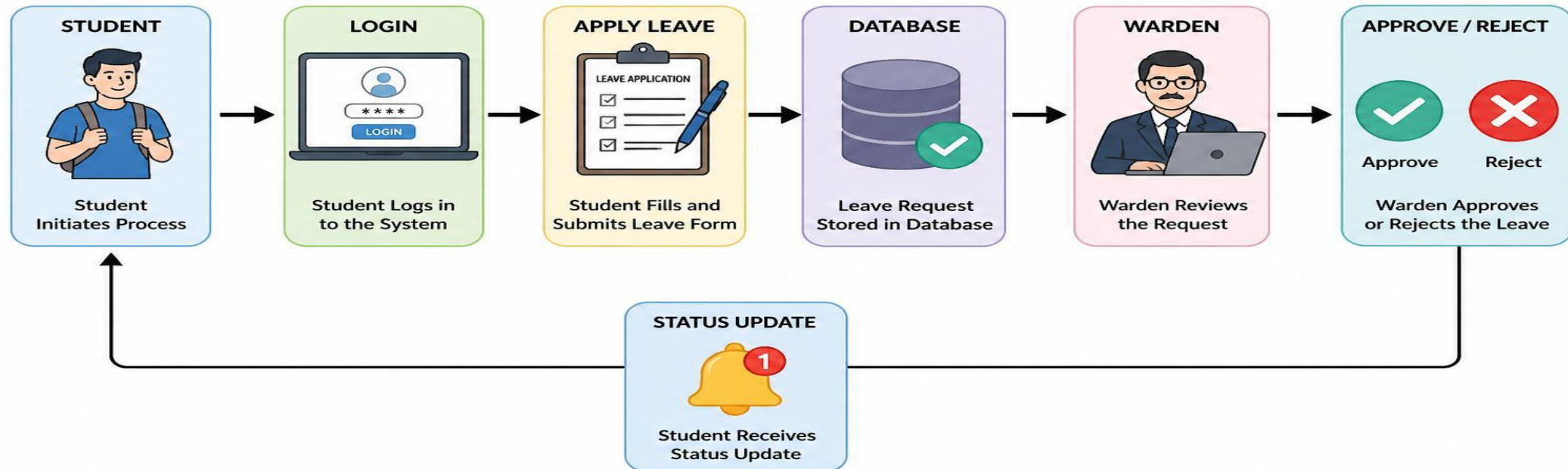
Address during Leave



# METHODOLOGY:

## 4. Working process

- **User Authentication**
  - Students and wardens log in using secure credentials.
  - Only authorized users can access the system.
- **Leave Application & Approval**
  - Students fill out an online leave form and submit it.
  - The warden reviews and approves or rejects the request.



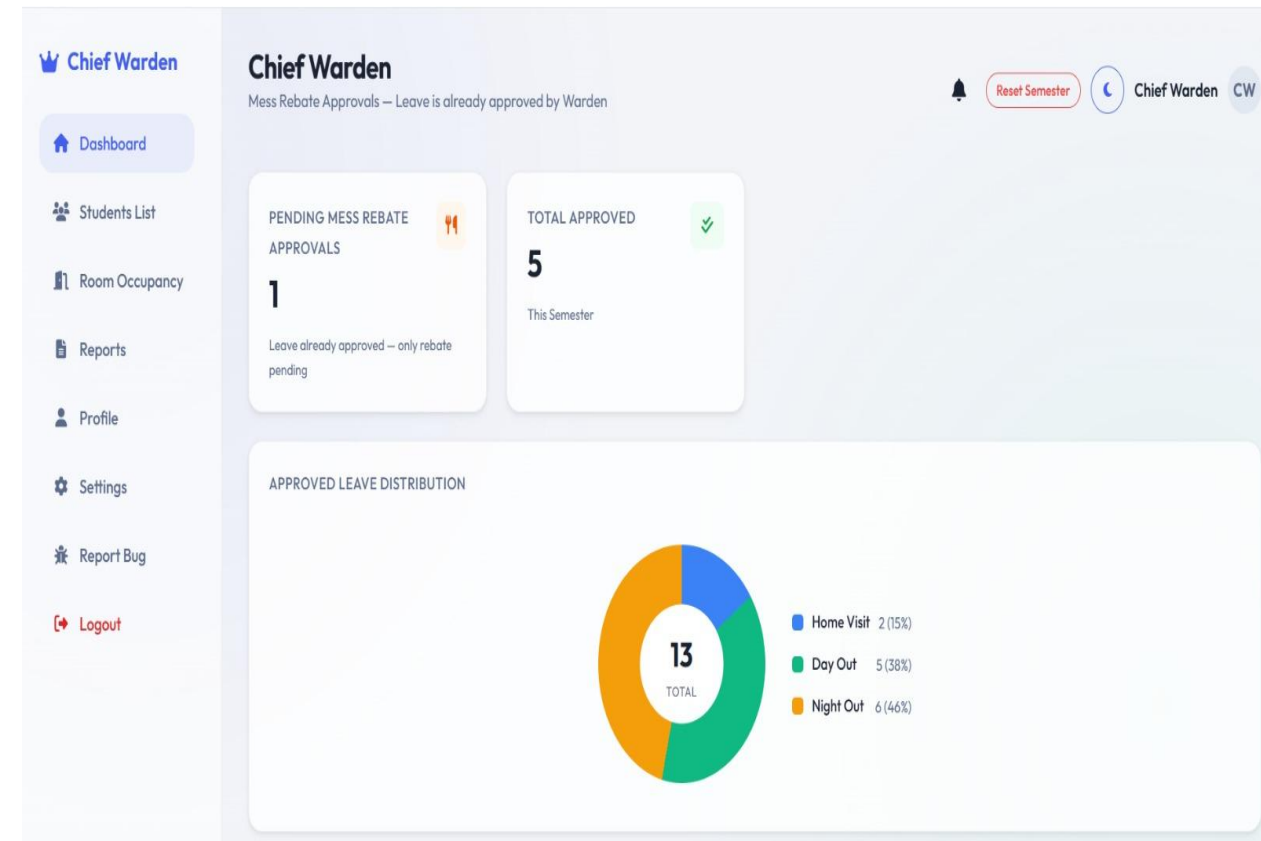
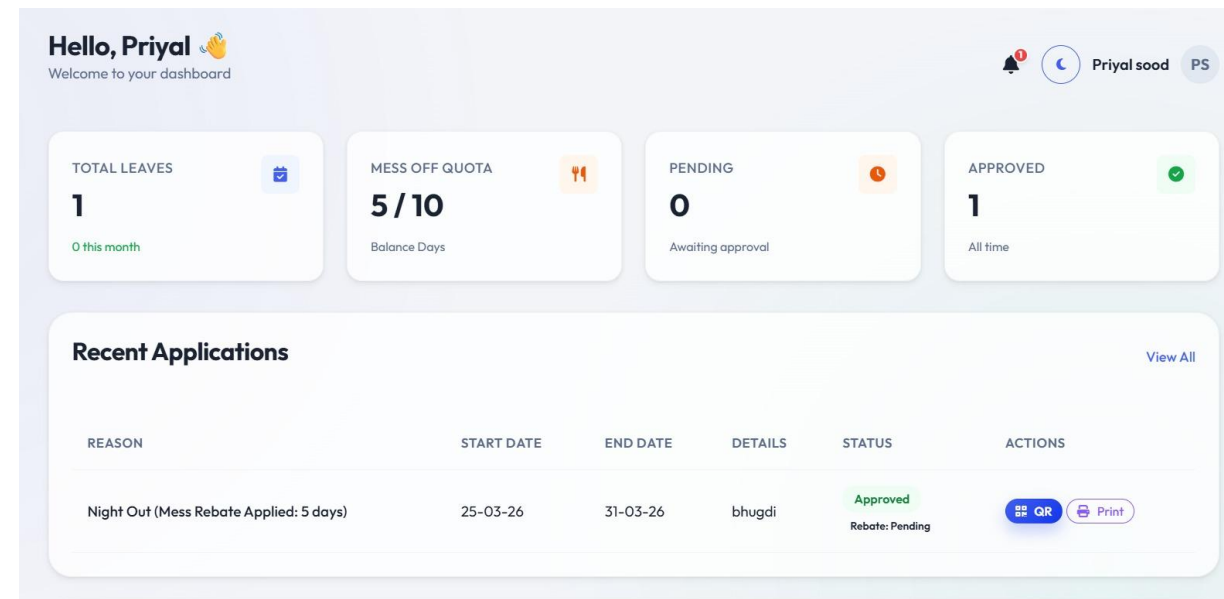
# RESULT & DISCUSSION:



## 1. Dashboard Implementation

We have successfully created all required dashboards for students and wardens .

These dashboards help in managing leave requests and viewing records easily.



# RESULT & DISCUSSION :



## 2. QR Code Feature

The system includes a QR code feature for quick verification of leave status .

This makes checking and validation faster and more efficient.

Gate Pass

Pass #119788

**Student:** Sakshi

**Status:** Approved

**From:** 25-03-26

**To:** 26-03-26

Close

## 3.Mess off feature

- Students can easily select and manage their mess off duration.

Mess Off Quota

Balance: 5 / 10 Days

Leave Type

Contact No.

Night Out

6230787072

From Date

To Date

2026-05-06

2026-05-12

☒ Apply for Mess Rebate (5 days eligible)

Note: Rebate is applicable only if leave duration (excluding travel days) exceeds 5 days.

☐ Apply for Mess Rebate (Quota Full)

Note: Rebate is applicable only if leave duration (excluding travel days) exceeds 5 days.



# RESULT & DISCUSSION :



## 3. Emergency leave feature:

- The system provides a special section for emergency leave requests.
- Students can quickly apply in urgent situations without delay.
- When a student applies for emergency leave, the request is first reviewed by the superintendent.
- Upon successful approval, the system automatically generates a unique QR code for that specific leave request.

☐

### ⚠ Emergency Leave

⚠ **For genuine emergencies only.** Checking this allows the Superintendent to **directly approve** your leave without waiting for Warden or Chief Warden review. Misuse of this option may result in disciplinary action. The Superintendent will be required to verify your situation by contacting your parent/guardian before approving.

Cancel

Submit Application

# CONCLUSION & FUTURE WORK:

- The proposed Hostel Leave Management System provides a digital and efficient solution for managing student leave and mess activities.
- It integrates features such as emergency leave approval with QR code generation, mess off management, and automated monitoring.
- The system reduces manual work, minimizes errors, and improves transparency between students and authorities.
- Overall, it enhances operational efficiency, ensures better record management, and provides a secure and user-friendly platform.

## FUTURE WORK:

- **Mobile Application Development**

The system can be extended into a mobile app for easier access.  
This will improve usability and real-time communication.

- **Biometric Integration**

Adding fingerprint or face recognition can enhance security.  
It will ensure accurate student identification.

- **Parental Notification**

In future enhancements, the system can include a Parental Notification Feature to keep parents or guardians informed about student activities such as leave approval, emergency leave, and mess off status.

**THANK YOU...**